Layer 2 Switching

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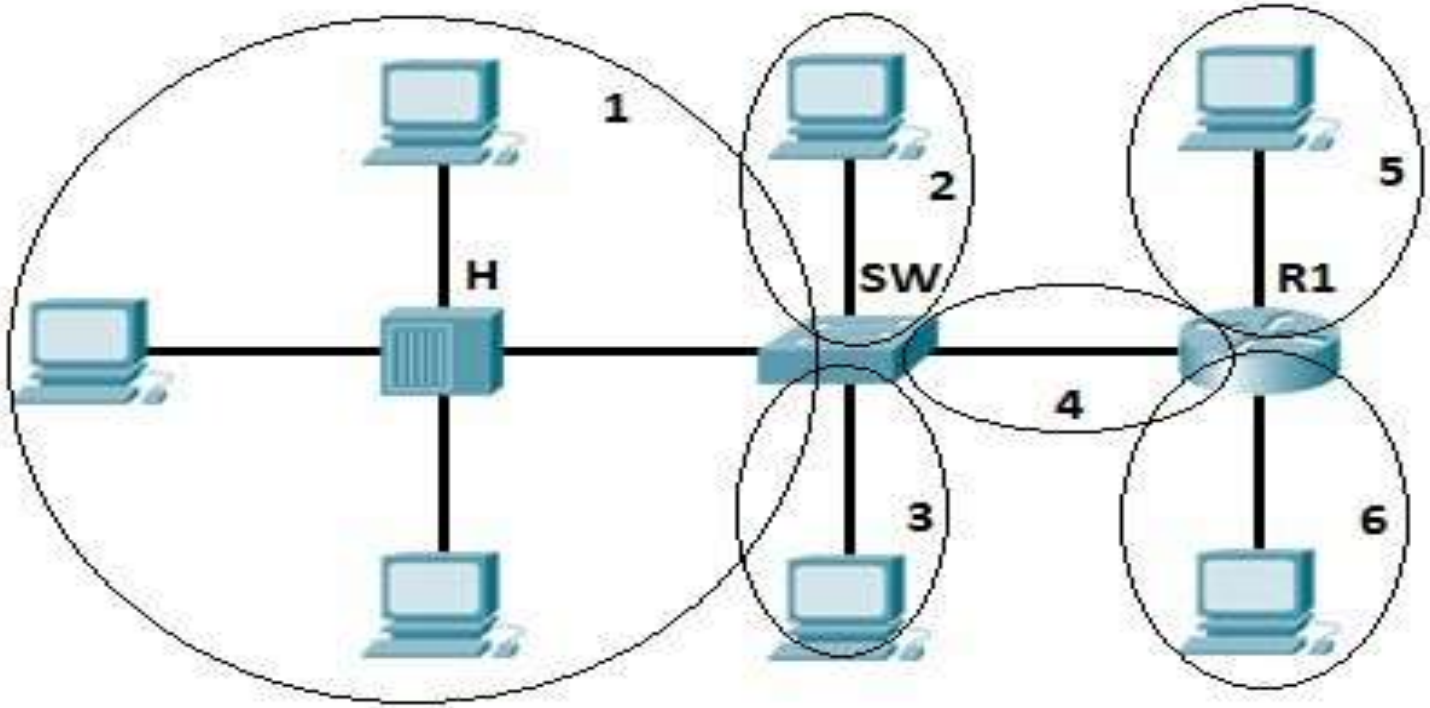
Layer 2 switching

- **Layer 2 switching** (or **Data Link layer switching**) is the process of using devices' MAC addresses to decide where to forward frames. Switches and bridges are used for Layer 2 switching. They break up one large collision domain into multiple smaller ones.

Collision Domain

- A collision domain is, as the name implies, the part of a network where packet collisions can occur. A collision occurs when two devices send a packet at the same time on the shared network segment. The packets collide and both devices must send the packets again, which reduces network efficiency. Collisions are often in a hub environment because each port on a hub is in the same collision domain. By contrast, each port on a bridge, a switch, or a router is in a separate collision domain.

Collision Domain

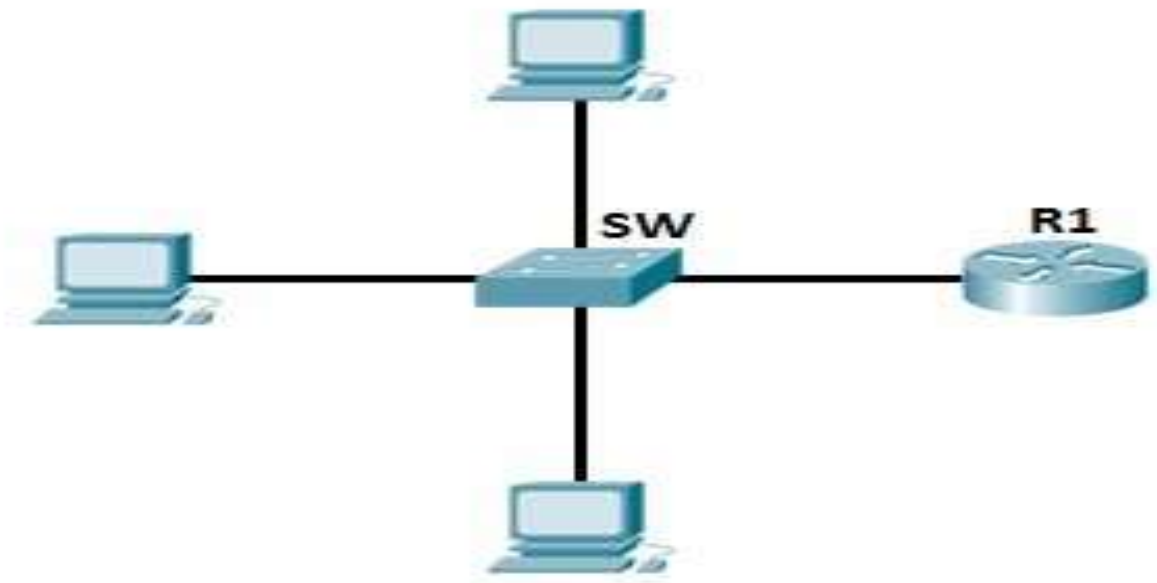


major advantages of Layer 2 switching

- fast hardware-based bridging (using ASICs chips)
- wire speed
- low latency
- low cost

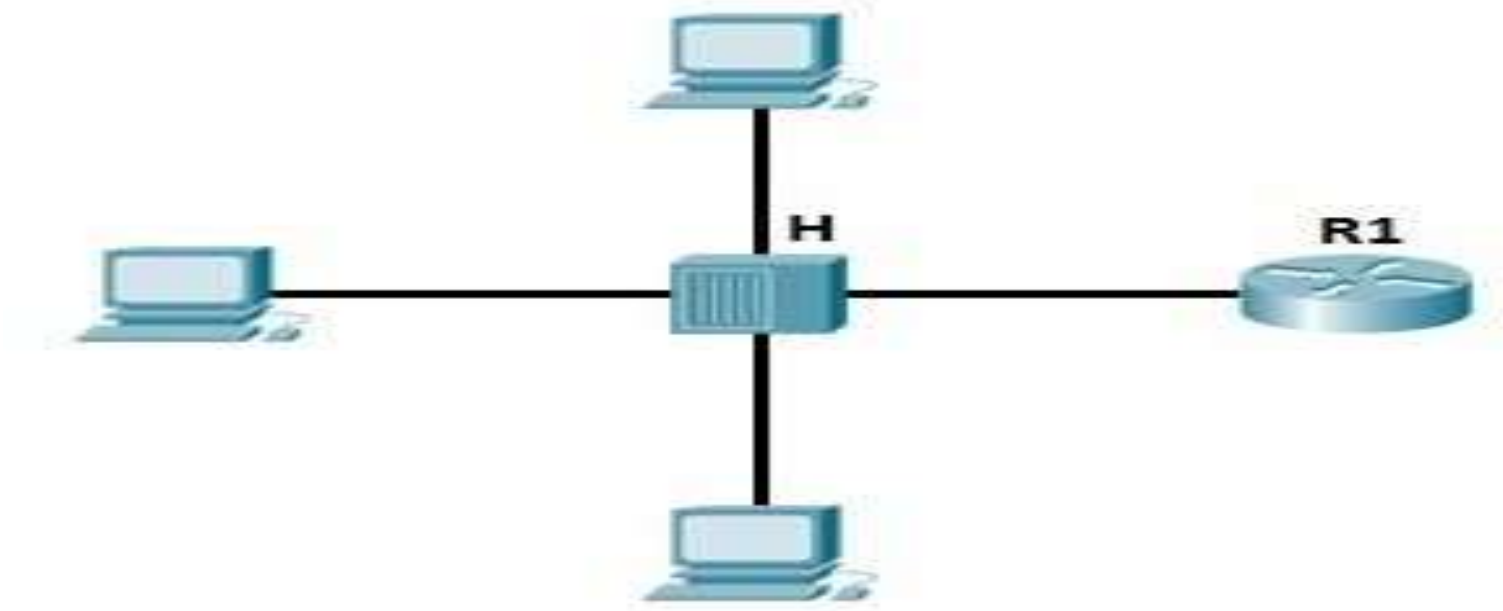
Typical LAN network

- the switch serves as a central device that connects all devices together:

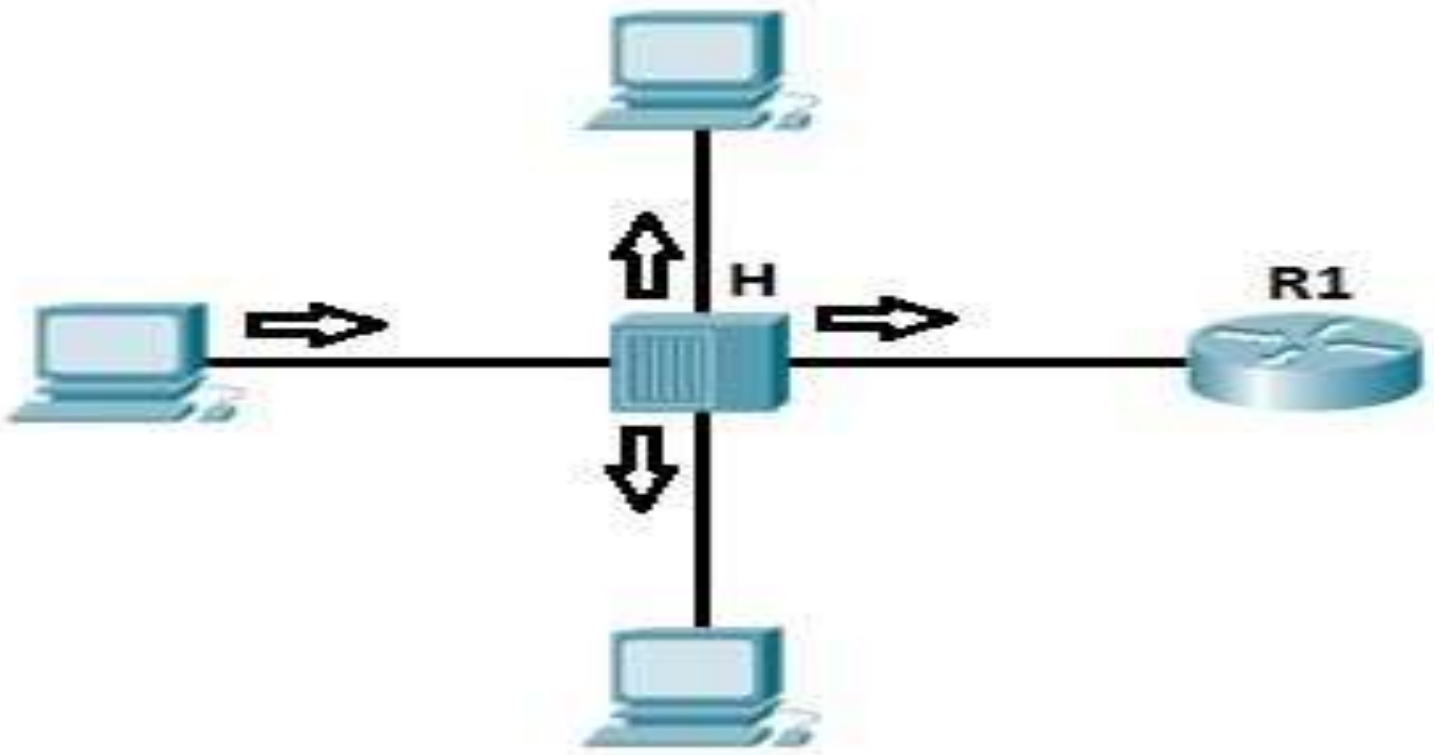


Differences between hubs and switches

- LAN in which all hosts connects to a hub

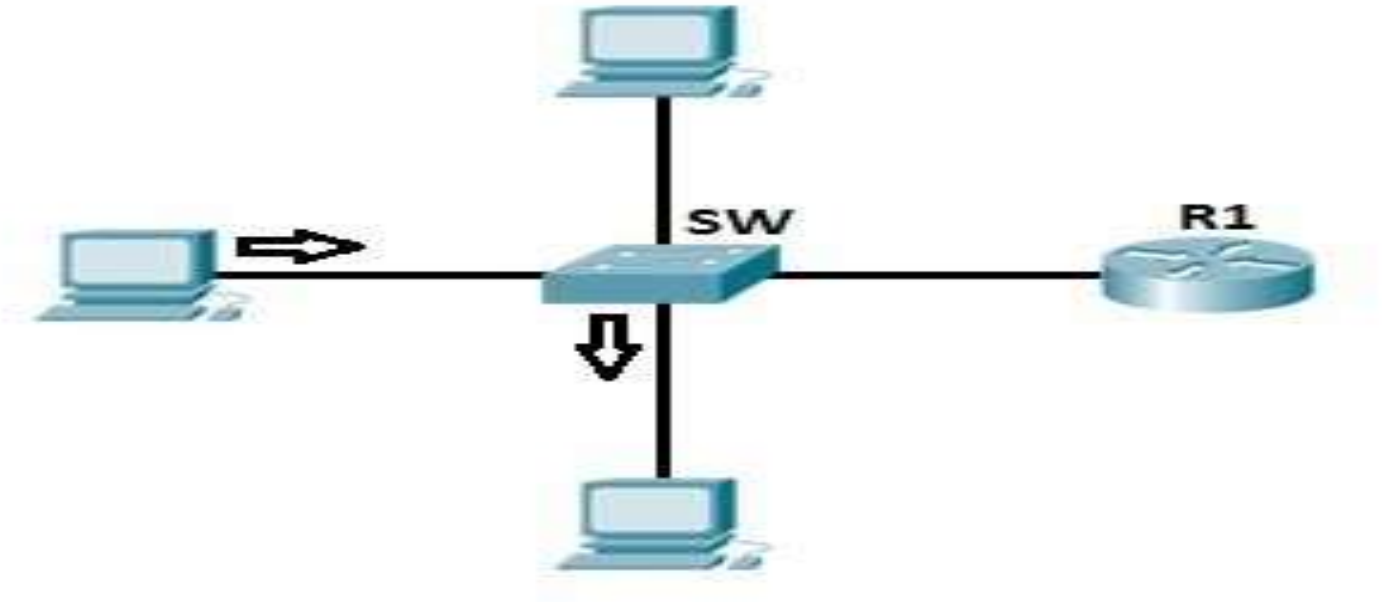


Differences between hubs and switches



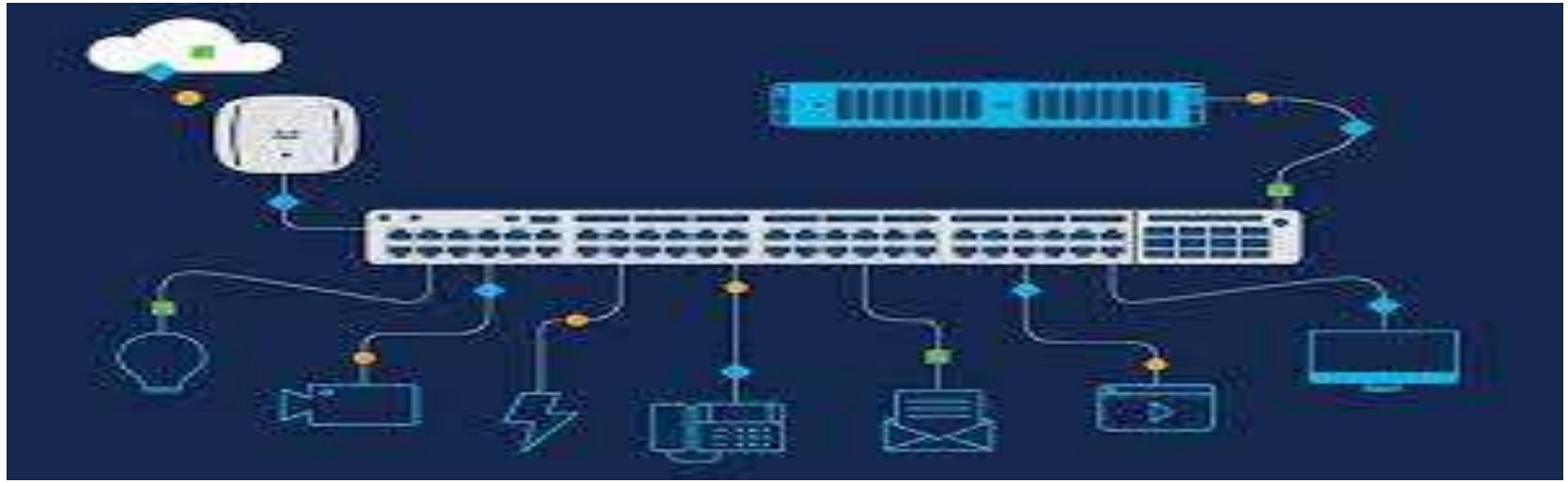
Differences between hubs and switches

Now consider the way the switches work. We have the same topology



How Switches Work

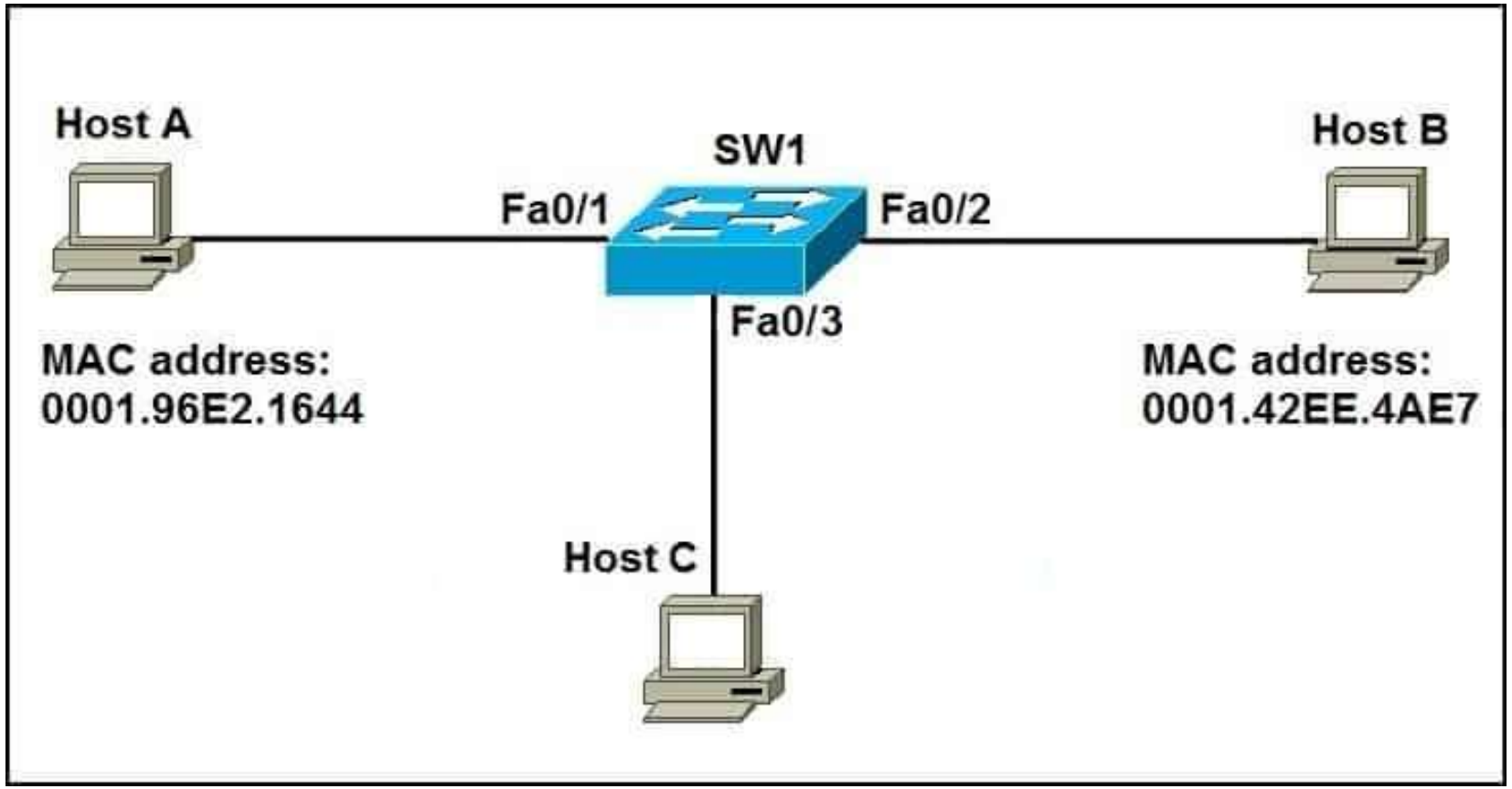
- Each network card has a unique identifier called a Media Access Control (MAC) address. This address is used in LANs for communication between devices on the same network segment. Devices that want to communicate need to know each other MAC addresses before sending out packets.



How Switches Work

- Switches usually perform these three functions in a LAN
  - **address learning** – switches learn MAC addresses by examining the source MAC address of each received frame.
  - **forward/filter decisions** – switches decide whether to forward or filter a frame, based on the destination MAC address.
  - **loop avoidance** – switches use Spanning Tree Protocol (STP) to prevent network loops while still permitting redundancy.

How Switches Work



What is Spanning Tree Protocol?

- **Spanning Tree Protocol (STP)** is a network protocol designed to prevent layer 2 loops. It is standardized as **IEEE 802.D** protocol.
  - Institute of Electrical and Electronics Engineers IEEE
  - EEE 802 is a collection of networking standards that cover the physical and data link layer specifications for technologies such as Ethernet and wireless.

Types of Network Switches

- Modular Network Switches
- Fixed-configuration Network Switches
- Managed Network Switches
- Unmanaged Network Switches
- Smart Ethernet Switches

Modular Switches

- Modular switches offer the flexibility to add expansion modules based on changing network requirements. These modules are specific to various applications like firewalls, wireless connectivity, or network analysis. They are more expensive than fixed-configuration switches for additional line cards, power supplies, or cooling fans and are typically used in large networks. Modular Ethernet network switches provide great flexibility but come at a higher cost.

Modular Switches

Fixed-configuration Network Switches

- Fixed-configuration switches, on the other hand, have a predetermined number of ports and are generally non-expandable, making them more cost-effective overall. Fixed-configuration switches can be categorized into three types. They are unmanaged switches, smart switches, and managed switches.

Fixed-configuration Network Switches

Nexus 5000 Portfolio

Nexus 56128P
- 2U
- 2 Expansion Slots (UP, 10G, 40G)
- Up to 96 10G Ports (48 UP)
- Up to 8 40G QSFP+ Ports

Nexus 5672UP
- 2U
- Up to 48 10G Ports (16 UP)
- Up to 6 40G QSFP+ Ports

Nexus 5624Q
- 1U
- 12 40G QSFP+ Ports

Nexus 5696Q
- 4U
- 8 Expansion Slots (UP, 10G, 40G, 100G)
- 96 40G QSFP+ Ports

Nexus 5648Q
- 2U
- 2 Expansion Slots (10G, 40G)
- 24 Fixed 40G QSFP+Ports

Nexus 5596T
- 2U
- 3 Expansion Slots (10GBaseT, 16UP)
- 32 Fixed 10GBaseT/16 SFP+

Nexus 5596UP
- 2U
- 3 Expansion Slots
- Up to 96 Ports (48 UP)

Nexus 5548P
- 1U
- 1 Expansion Slot
- Up to 48 Ports 10G (8 FC)

Nexus 5548UP
- 1U
- 1 Expansion Slot
- Up to 48 Ports 10G (16 UP)

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Managed Network Switches

- Managed switches, among the fixed-configuration switches, offer the most comprehensive features. They can provide excellent application experience, high levels of security, easy management, and scalability for future networks. As a result, they are commonly deployed as aggregation/access switches in large networks or as core switches in smaller networks. Managed switches are the most expensive option among fixed-configuration switches.

Unmanaged Network Switches

- Unmanaged switches are designed to be plug-and-play and no configuration is needed. They are most suitable when simple switching and connectivity are required, commonly found in home networks or environments where only a few ports are needed, such as desks, labs, or conference rooms. Some unmanaged switches may offer limited advanced capabilities. However, as the name implies, these switches generally can not be modified or managed.

Unmanaged Network Switches

S1900-24T, 24 x Gigabit RJ45, L2 Unmanaged Switch

Smart Ethernet Switches

- Smart switches provide management, segmentation, quality of service, and security capabilities, making them a cost-effective alternative to modular switches. However, they are not as scalable as managed switches. Smart switches are often deployed at the edge of large networks, while managed switches are used in the core. They are also suitable for smaller networks or those with low complexity. T

Smart Ethernet Switches

S2805S-24TF, 24 x Gigabit RJ45, L2 Smart Managed Switch

Realtek RTL8382M | LACP | DHCP |